The future of Operating Systems is Micro

Operating systems are huuuuge but do they have to be? Smaller OSes mean less surface-area for bugs to manifest or attackers to exploit. What's the catch, and how can we achieve this?

Level

Intermediate-Advanced

Details

Outlines

- This presentation walks through some of the many advantages in security, stability and modularity of microkernels.
- We discuss how our Microkernel-based Unix-like Operating System, Redox OS, can bring these to life and how this is achieved from a developer's perspective.
- Looking at some code to gain insight into how simple system services can be and hence how they can be so stable, despite performing relatively complex tasks. We cover:
  - device drivers,
  - process control
  - memory management
  - the anatomy of system calls
  - how schemes form the glue between the most critical parts of the OS

Outcomes

- We aim to bring the advances of the Rust programming languages and the benefits it brings to the attention of those interested in its broader applications in systems and services, as well as to end-users in the form of stable and reliable software.
- To encourage a philosophy of modular and repeatable systems in the search of stability and reliability with Rust.

Audience

- OS Devs, old and new. People interested in the next generation of systems, devoid of historical artefacts and nuances.
- Curious minds, eager to explore Rust and its many, many uses.

Pitch

With the help of The Redox Developers, I wish to introduce our efforts.. I've had the honour and pleasure of watching the development of the operating system unfold and working with the team. Consequently, I consider myself well-versed in the technical aspects of some of its most core infrastructure, and can see clearly the need to replace old and outdated technologies with new and modern ones. Technologies which form not the upper-bound of development, but the lower.

Speaker

Jacob Schneider, The Redox Developers

jacob.schneider.2003@outlook.com

Biography

I'm Jacob, I have a strong passion for low-level systems and operating systems and have been part of the Redox developers since December 2022

Shirt

Unfitted, Medium